1. Why is the Moon's gravitational pull on the Earth's water twice as strong as the sun?

   Although the Moon weighs a lot less than the Sun, the Moon is much closer to the Earth that the gravitational pull on our water is twice as strong as that of the sun.

2. What causes tides?
   Tides are caused by the gravitational interaction between the Earth and the Moon.

3. Explain how gravity affects the Earth and the Moon.
   Through gravity, the Earth pulls on the Moon to hold it in its orbit. The Moon also tries to pull at anything on the Earth to bring it closer. The Earth can hold onto everything except water.

4. What effect does the gravitational attraction of the moon have on the Earth's oceans?
   The gravitational attraction of the moon causes the oceans to bulge out in the direction of the moon. Another bulge occurs on the opposite side, since the Earth is also being pulled towards the moon (and away from the water on the far side).

5. Why do we have two tides each day?
   The earth is rotating while this is happening, so two tides occur each day.

6. High tide occurs when the gravitational pull of the Moon causes the ocean to bulge out in the direction of the Moon. This is what we refer to as a high tide.

7. Why is there a stronger gravitational pull during a full or new moon?
   During these phases, the Sun, Moon and Earth are aligned and their combined gravity causes very high and very low tides. These can be referred to as spring tides.

8. What does the 'spring' in spring tide mean?
   Spring as in jump, not spring as in the season after winter.

9. Why do we have smaller tides during the first and third quarters of the Moon's phases?
   During these phases, the Sun and Moon are at right angles to Earth and the gravitational forces cancel each other out, hence smaller tides occur. These are referred to as neap tides.

10. During a **spring** tide both the moon and sun pull on the ocean water. The high tide is **more** extreme.

    During a **neap** tide, the sun and the moon pull the opposite of each other. High tide is **less** extreme.

11. True or False
    It takes the moon one month to orbit the earth. True 29.5 days
    The moon rotates in the opposite direction to the earth. False
    The moon rotates at a much slower speed than Earth. True

12. The tidal cycle is not 24 hours long, but 24 hours and 50 minutes long because it takes an extra 50 minutes each day for the moon to be in the same location, relative to a point on earth, to the previous day. This is why high and low tides are approximately 50 minutes later each day.